Daily Work Log — 2 November 2026

Lifecycle Automation in PowerShell (Step 8b) · Live Leaver Deprovisioning · Hire Dates · Azure Clean-up Check · Documentation

What was done today, why, and the evidence. Tenant ID, object IDs and the tenant domain are masked because this repository is public. Terminal screenshots come from read-only commands and the evidence files saved today; Figure 5 is my own PowerShell window.

Project	Healthcare IAM Operations Simulation — Horizon Valley Medical Center (HVMC)
Environment	Microsoft Entra ID (Free tier) · PowerShell 7.6.9 + Microsoft Graph SDK · GitHub repository dae-6-month-Sabi
Prepared By	Sabitha S Karanavar
Date	November 2, 2026
Cost	$0 — Entra ID Free features only; the Azure subscription was checked, not used

Summary of the day

#	Work	Reference	Result
1	Step 8b: joiner/mover/leaver automation in PowerShell — HR feed, daily leaver check, shared module	HVMC-JML-204/205/206 · finding F10	**Done, 16/16 steps ok**
2	Leaver Victor Mensah removed by the daily leaver check, run by me after his shift ended	HVMC-JML-206	**Done, 5 minutes after leave time**
3	Hire dates recorded for all 30 bulk-created staff accounts	HVMC-IAM-CHG-110 · finding F11	**Done, 30/30**
4	IAM report updated: Step 9 password reset (v1.3), Step 8b (v1.4), leaver result and hire dates (v1.5)	commits e92e3f8, a76e6c0	**Pushed**
5	Daily work log for 29 October with screenshots	commit 1bd0c64	**Pushed**
6	Azure subscription checked: what must be removed before Pay-As-You-Go	Key Vault preparation	Read-only, nothing changed
7	Documentation audit, PowerShell PATH check, Claude Code chat retention set to 365 days	—	**Done**

1. Step 8b — Lifecycle automation in PowerShell

Why. On 20 October the joiner, mover and leaver changes were separate scripts that I started by hand, one event at a time (finding F10). Nothing checked that a leave date was acted on. A leaver whose last day passes unnoticed keeps access to patient data. Step 8b makes the process repeatable and driven by HR, using **PowerShell 7 with the Microsoft Graph SDK**, the standard tooling for Microsoft identity administration.

File	What it does
Invoke-HvmcLifecycle.ps1	Reads the HR feed (CSV: ticket, Join / Move / Leave, employee ID, department, title, manager, date, optional end-of-shift time) and plans each event from the RBAC v2 matrix
Invoke-HvmcLeaverCheck.ps1	Daily safety net: removes access once a leave date has passed, warns 14 days ahead, flags contractors with no end date
Set-HvmcHireDates.ps1	Records hire dates from an HR extract (CHG-110, section 3)
HVMC.Lifecycle.psm1	Shared functions: sign-in, desired groups, user state, plans, evidence
config/rbac_matrix.json	Exported from step2_rbac_v2/rbac_v2.py, so the Python matrix stays the single source of truth

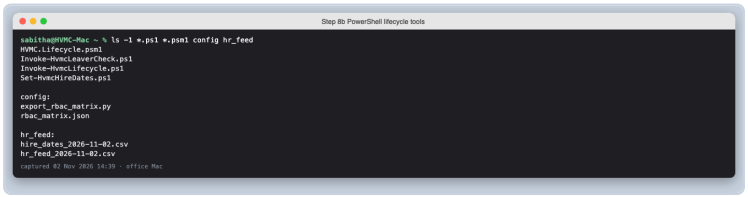


Figure 1. The Step 8b tools: three scripts, the shared module, the RBAC config and the HR files.

Safety controls. Every script shows a dry run first and changes nothing without `-Apply`. Running a script twice is safe: it checks the current state and reports “already done”. Accounts with an admin role are never touched, and every row is checked before use. On a transfer, the old role’s groups are removed *before* the new role’s are added (separation of duties). New passwords are random and never shown.

The run. The HR feed held three new events: joiner Ethan Morales (Lab Technician), mover Jason Park (IT to Security Officer), and leaver Victor Mensah. A fourth row repeated Olivia Bennett’s joiner from 20 October to prove a re-run is harmless.

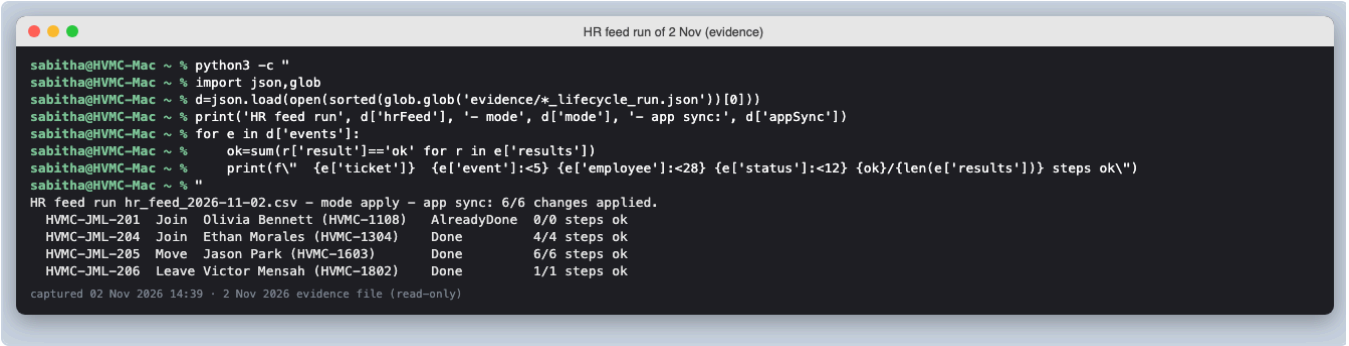


Figure 2. The applied run: all steps succeeded, the existing joiner was skipped, and the application sync made 6 matching changes.

Employee	Before	After
Ethan Morales (joiner)	No account	Laboratory; 2 Lab groups; EHR (read) and Laboratory Information System
Jason Park (mover)	IT; IT Admin Console, Scheduling admin, VPN	Security; Security Monitoring Portal only — IT admin access removed first
Victor Mensah (leaver)	Reception; EHR (read), Appointment Scheduling	Leave time recorded; removed by the leaver check (section 2)

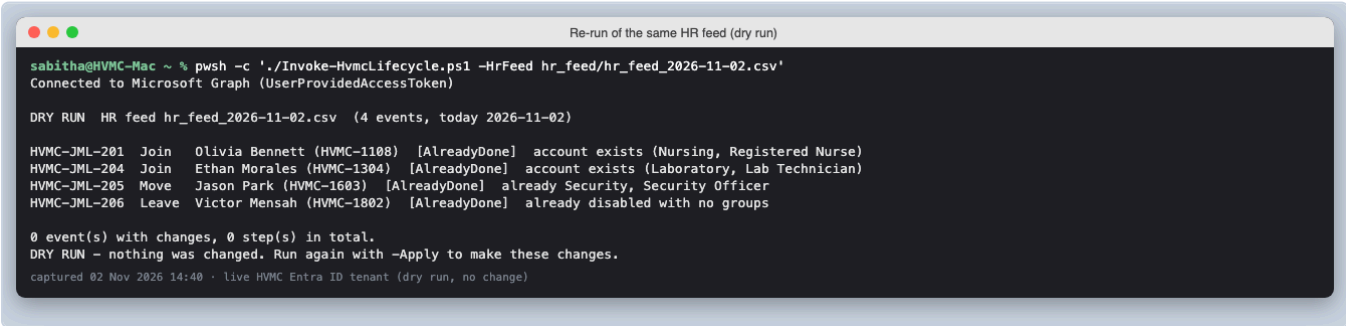


Figure 3. The same feed run again as a dry run: every event is “already done”, so nothing would change.

Defects found by testing and fixed: the first run saved its “after” snapshot before the application sync, and the simulated-date option showed local time labelled as UTC. Both were fixed and the correct state was re-recorded.

2. Leaver: Victor Mensah (HVMC-JML-206)

The HR feed first gave Victor's last day as 6 November. HR then moved it to **today, ending at 14:30 EST (19:30 UTC)**, the end of his shift. A leave date normally ends at midnight UTC, so I added an optional **end-of-shift time** column to the feed. The feed only recorded the new time, and Victor kept his access while still working. A leaver check run before 14:30 gave a notice and changed nothing, which is correct.

After 14:30 I ran the leaver check myself in PowerShell:

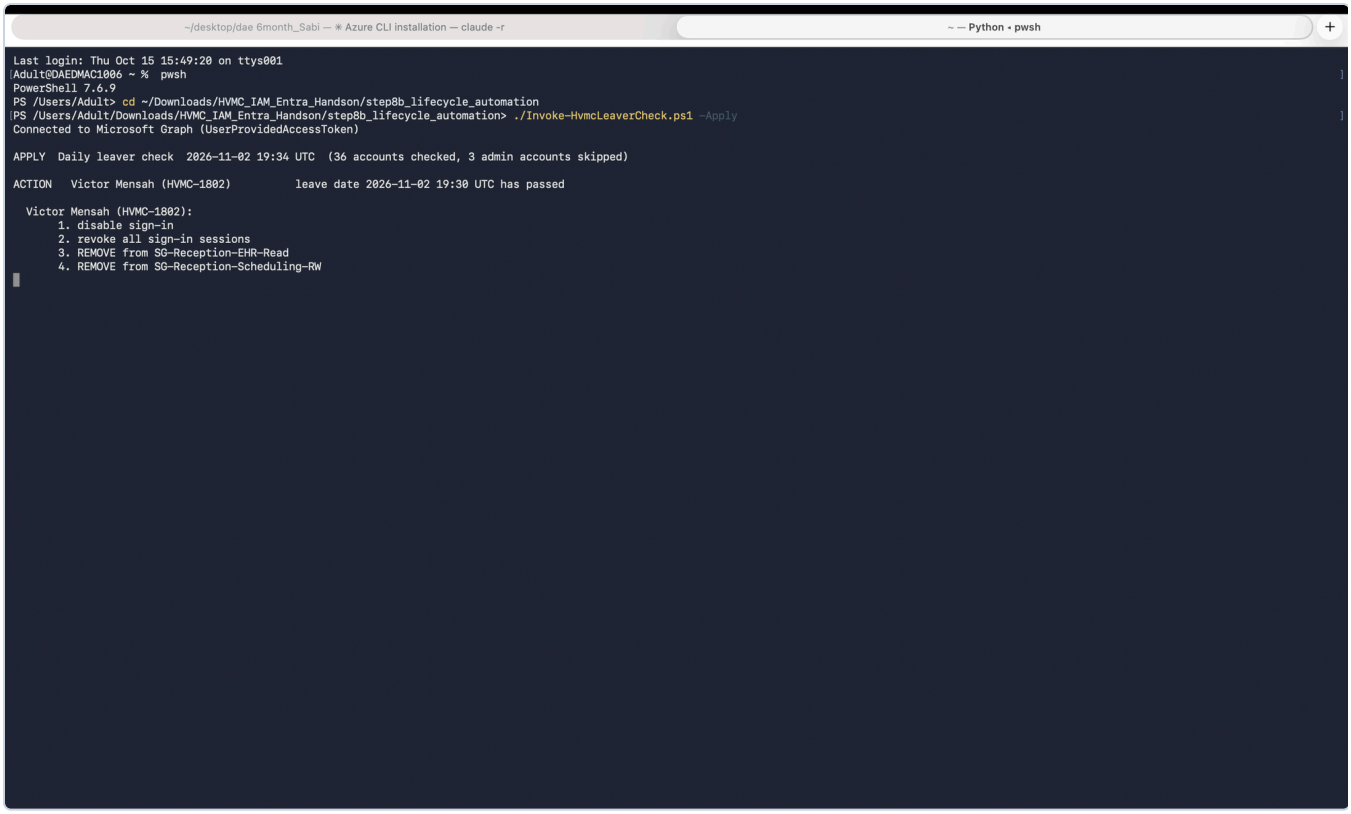


Figure 4. My PowerShell window: the leaver check with -Apply finds that Victor's leave time has passed and lists the four steps.

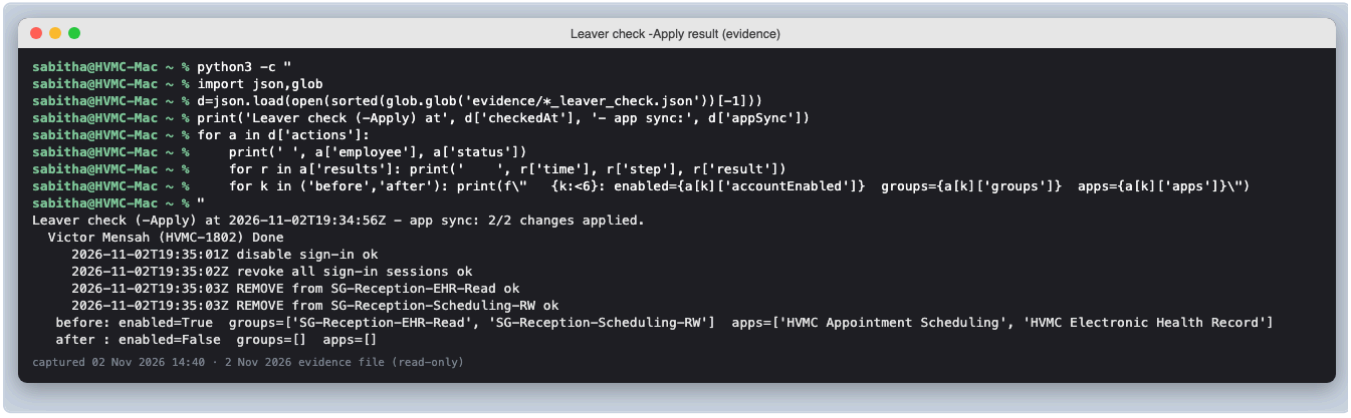


Figure 5. Result: 4/4 steps ok within a minute, then the application sync removed his two applications. Before and after are recorded.

Check	Result
Time to remove access	**5 minutes** after his shift ended (19:30 → 19:35 UTC)
Account	Sign-in disabled; all sessions revoked
Groups and applications	Removed from 2 Reception groups; EHR and Appointment Scheduling removed
Check run again	No leavers due — nothing left to do (Figure 6)
Whole tenant	30/30 active staff match the RBAC design; 0 application assignments pending (Figure 7)

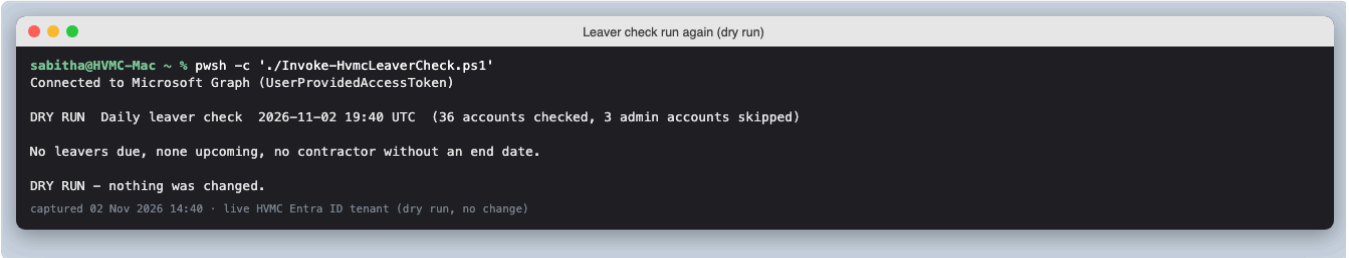


Figure 6. The leaver check run again: no leavers due, none upcoming, no contractor without an end date.

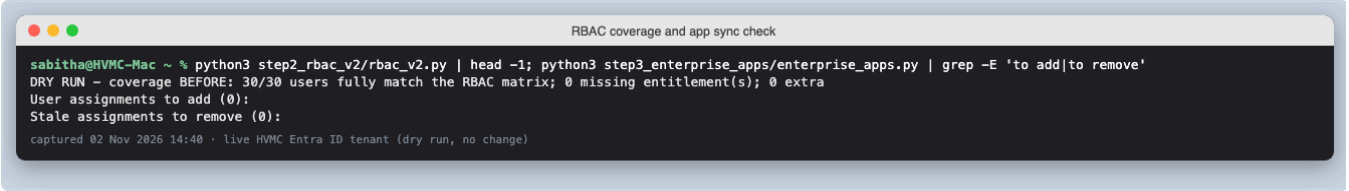


Figure 7. Every active staff member matches the RBAC v2 matrix, and no application assignment is missing or stale.

3. Hire dates (HVMC-IAM-CHG-110, finding F11)

Why. While checking Victor's record I found that his **join day was empty**. The 30 staff accounts were bulk-created without hire dates; only Olivia and Ethan, who came through the joiner process, had one. Access reviews and audits use the hire date to see how long someone has held access.

How. `Set-HvmcHireDates.ps1` loaded an HR extract and recorded **30 of 30** hire dates, re-reading each one to confirm. It only fills empty dates, never overwrites an existing one, and skips admin accounts. The dates are **simulated lab data** from a fixed rule: department leads in 2014, later staff in employee-ID order, always a Monday. Victor's hire date is 6 June 2016, so he left after about ten years.

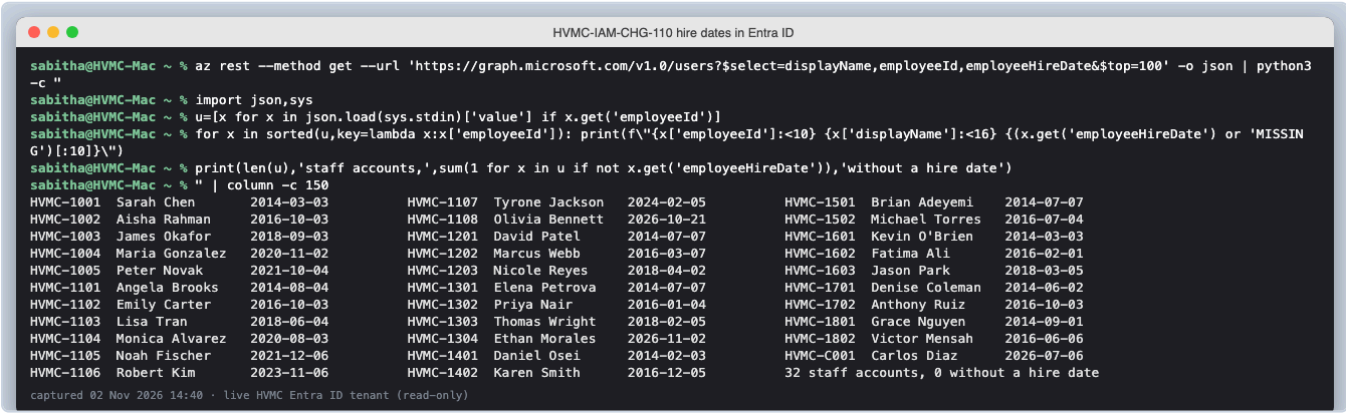


Figure 8. All 32 staff accounts now have a hire date; Olivia Bennett and Ethan Morales got theirs through the joiner process.

4. Azure clean-up check for Key Vault

Key Vault needs an active subscription. Before upgrading the disabled free trial to Pay-As-You-Go, I listed what is left from the SecOps and Cloud Infrastructure labs. It is a read-only check, and nothing was deleted.

Would cost money if upgraded	Free
NAT gateway, Standard load balancer, 2 public IPs, Log Analytics + Sentinel, flow log + Traffic Analytics, 2 storage accounts, activity-log export, Defender for Storage and Resource Manager (about 11 trial days left)	3 NSGs, virtual network, 9 alert rules, action group, workbook, data collection rules, Event Grid topics, Network Watcher, $10 budget

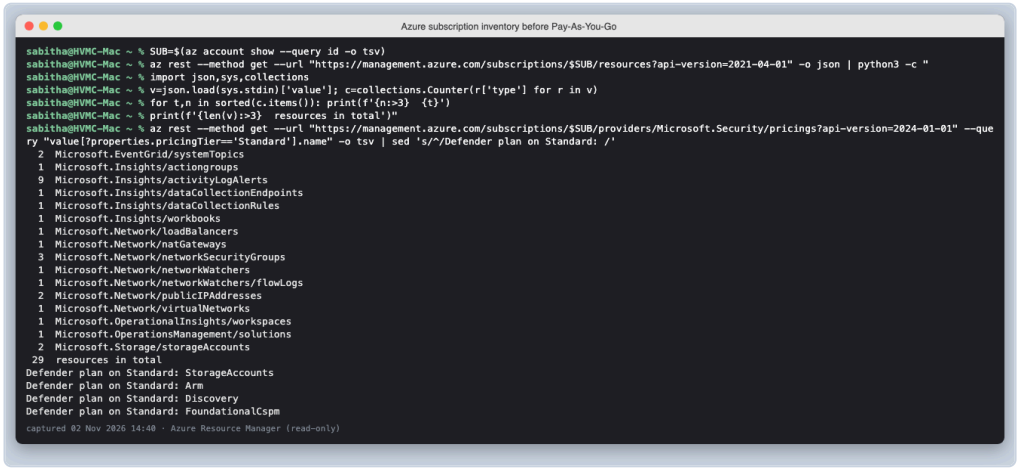


Figure 9. 29 resources remain, and four Defender plans are on Standard. Estimated $90–130 a month if left running after an upgrade.

The options are to delete the resource group `HVMC-CloudSecurity-RG` and reset the Defender plans before upgrading, or to use Azure for Students instead. I have not decided yet.

5. Documentation and housekeeping

- **IAM report** updated three times: Step 9 password reset (v1.3), Step 8b (v1.4), then Victor's real deprovisioning and the hire dates (v1.5).
- **Daily work log for 29 October** written and pushed with seven screenshots.
- **Documentation audit:** every step (1–9 and 8b) is documented and on GitHub. Five other documents (three weekly status decks, the DC build PDF and the project story) stay on my Mac by my choice.
- **Open item found:** two scripts pushed on 20 October (identity_hygiene.py, jml.py) contain the tenant domain. They are to be masked in a later commit.
- **Tools:** confirmed PowerShell starts from Terminal (PATH already set). Claude Code chat retention was raised from 30 to 365 days so project chats stay available.

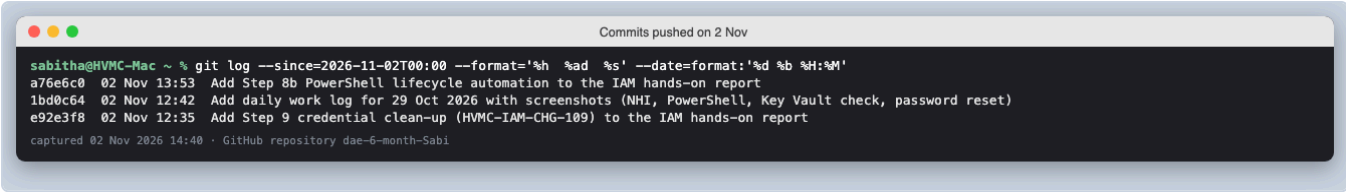


Figure 10. Commits pushed today before this log.

6. What I learned

- **PowerShell runs on a Mac.** PowerShell 7 is cross-platform, and the Graph modules talk to the cloud over HTTPS. Only the Windows-only modules, such as ActiveDirectory, need Windows.
- **A leave date needs a time.** “Last day 2 November” can mean midnight or the end of a shift. Removing access before someone stops working is as wrong as removing it late.
- **Automation must be safe to run twice.** Checking the current state first turns a repeated run into “already done” instead of duplicate accounts.
- **Test that it works, then test it again.** Testing found two defects before anyone relied on the evidence.
- **Data gaps show up in real work.** The missing hire dates were found by looking at one leaver’s record.

7. Next steps

Action	When
Choose the Azure option (Azure for Students, or clean up and Pay-As-You-Go) for Key Vault	Next
Mask the tenant domain in identity_hygiene.py and jml.py	Next
Schedule the daily leaver check (Azure Automation runbook with a managed identity)	After the subscription
Regenerate the SerpAPI key at serpapi.com	As soon as possible
Quarterly break-glass sign-in test	About 28 Jan 2027
Access review HVMC-AR-2027-Q1	January 2027